凑微分专项训练卷

(1) $\int f(ax+b)dx = \frac{1}{a}\int f(ax+b)d(ax+b)$

1. $\int e^{2x+1}dx =$ _____
2. $\int \sin(3x-2)dx =$ _____
3. $\int (4x+5)^6 dx =$ _____
4. $\int \frac{1}{5x+3}dx =$ _____
5. $\int \sec^2(2x)dx =$ _____

(2) $\int x^m f(ax^{m+1}+b)dx = \frac{1}{(m+1)a}\int f(ax^{m+1}+b)d(ax^{m+1}+b) \quad (m \neq -1)$

1. $\int xe^{x^2+1}dx =$ _____
2. $\int x^2\cos(2x^3-1)dx =$ _____
3. $\int x^3\sqrt{3x^4+2}dx =$ _____
4. $\int \frac{x}{4x^2+5}dx =$ _____
5. $\int x^4\sin(x^5)dx =$ _____

(3) $\int f(\sqrt{x})\frac{dx}{\sqrt{x}} = 2\int f(\sqrt{x})d(\sqrt{x})$

1. $\int \frac{\sin(\sqrt{x})}{\sqrt{x}}dx =$ _____
2. $\int \frac{e^{\sqrt{x}}}{\sqrt{x}}dx =$ _____
3. $\int \frac{\cos(\sqrt{x})}{\sqrt{x}}dx =$ _____
4. $\int \frac{1}{\sqrt{x}(1+\sqrt{x})}dx =$ _____
5. $\int \frac{\ln(\sqrt{x})}{\sqrt{x}}dx =$ _____

(4) $\int f(e^x)e^x dx = \int f(e^x)d(e^x)$

1. $\int e^x\cos(e^x)dx =$ _____
2. $\int e^x\sin(e^x)dx =$ _____
3. $\int \frac{e^x}{1+e^{2x}}dx =$ _____
4. $\int e^x\sqrt{1+e^x}dx =$ _____
5. $\int e^x\ln(e^x)dx =$ _____

(5) $\int f(\ln x)\frac{1}{x}dx = \int f(\ln x)d(\ln x)$

1. $\int \frac{\cos(\ln x)}{x}dx =$ _____
2. $\int \frac{\sin(\ln x)}{x}dx =$ _____
3. $\int \frac{1}{x(1+\ln^2 x)}dx =$ _____
4. $\int \frac{\ln^2 x}{x}dx =$ _____
5. $\int \frac{e^{\ln x}}{x}dx =$ _____

(6) $\int f(\sin x)\cos x dx = \int f(\sin x) d(\sin x)$

1. $\int \sin^2 x \cos x dx =$ _____
2. $\int e^{\sin x}\cos x dx =$ _____
3. $\int \cos(\sin x)\cos x dx =$ _____
4. $\int \frac{\cos x}{1+\sin^2 x} dx =$ _____
5. $\int \sqrt{\sin x}\cos x dx =$ _____

(7) $\int f(\cos x)\sin x dx = -\int f(\cos x) d(\cos x)$

1. $\int \cos^3 x \sin x dx =$ _____
2. $\int e^{\cos x}\sin x dx =$ _____
3. $\int \sin(\cos x)\sin x dx =$ _____
4. $\int \frac{\sin x}{1+\cos^2 x} dx =$ _____
5. $\int \sqrt{\cos x}\sin x dx =$ _____

(8) $\int f(\tan x)\frac{1}{\cos^2 x} dx = \int f(\tan x) d(\tan x)$

1. $\int \tan^3 x \frac{1}{\cos^2 x} dx =$ _____
2. $\int e^{\tan x}\frac{1}{\cos^2 x} dx =$ _____
3. $\int \sin(\tan x)\frac{1}{\cos^2 x} dx =$ _____
4. $\int \frac{1}{1+\tan^2 x}\frac{1}{\cos^2 x} dx =$ _____
5. $\int \sqrt{\tan x}\frac{1}{\cos^2 x} dx =$ _____

(9) $\int f(\arcsin x)\frac{1}{\sqrt{1-x^2}} dx = \int f(\arcsin x) d(\arcsin x)$

1. $\int \frac{(\arcsin x)^2}{\sqrt{1-x^2}} dx =$ _____
2. $\int \frac{e^{\arcsin x}}{\sqrt{1-x^2}} dx =$ _____
3. $\int \frac{\cos(\arcsin x)}{\sqrt{1-x^2}} dx =$ _____
4. $\int \frac{1}{\arcsin x\sqrt{1-x^2}} dx =$ _____
5. $\int \frac{\sqrt{\arcsin x}}{\sqrt{1-x^2}} dx =$ _____

(10) $\int f(\arctan x)\frac{1}{1+x^2} dx = \int f(\arctan x) d(\arctan x)$

1. $\int \frac{(\arctan x)^3}{1+x^2} dx =$ _____
2. $\int \frac{e^{\arctan x}}{1+x^2} dx =$ _____
3. $\int \frac{\sin(\arctan x)}{1+x^2} dx =$ _____
4. $\int \frac{1}{(1+(\arctan x)^2)(1+x^2)} dx =$ _____
5. $\int \frac{\sqrt{\arctan x}}{1+x^2} dx =$ _____